

Steffensen

Newton

Equations:

$$f(x) = x^2 - 2 = 0$$

Lagrange Mean Value Theorem: $f(x)$ has to be very smooth, $f'(x)$ is **continuous**

$$\exists c \in [a, b], \quad s.t. f'(c) = \frac{f(a) - f(b)}{a - b}$$

Newton, assume α is the zero point ($f(\alpha) = 0$)

$$f(x) \approx f(\alpha) + f'(x)(x - \alpha) = f'(x)(x - \alpha)$$

Simplify it:

$$\alpha \approx x - \frac{f(x)}{f'(x)}$$

Iteration:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = \varphi(x_k)$$

Square root iteration:

$$x_{k+1} = x_k - \frac{x_k^2 - 2}{2x_k} = \frac{1}{2} \left(x_k + \frac{2}{x_k} \right)$$

Another way

Equation:

$$x = \frac{2}{x}$$

Implies, $c \in (0, 1)$

$$x = cx + (1 - c)x = cx + (1 - c)\frac{2}{x}$$

Iteration:

$$x_{k+1} = cx_k + (1 - c)x_k = cx_k + (1 - c)\frac{2}{x_k}$$

Proof of Convergence:

- Take the subtraction:

$$|x_{k+1} - x_k| = \left| c - \frac{2 - 2c}{x_k x_{k-1}} \right| |x_k - x_{k-1}|$$

- Analyse the coefficient:

$$x_0 > 0 \implies x_{k+1} = cx_k + (1 - c)\frac{2}{x_k} \geq 2\sqrt{2c(1 - c)}$$

Thus:

$$x_{k+1}x_k \geq 8c(1 - c) \implies \delta = c - \frac{2 - 2c}{x_k x_{k-1}} \geq c - \frac{1}{4c}$$

On the other hand, $c \in (0, 1)$, thus

$$\delta < c < 1$$

If $c > \frac{\sqrt{2} - 1}{2} \approx 0.207106$, then

$$-1 < c - \frac{1}{4c} \leq c - \frac{2 - 2c}{x_k x_{k-1}} = \delta < c < 1$$

Thus $|\delta| < 1$, we have:

$$|x_{k+1} - x_k| = |\delta| |x_k - x_{k-1}| = |\delta|^2 |x_{k-1} - x_{k-2}| = \cdots |\delta|^k |x_1 - x_0|$$

Which means:

$$\lim_{k \rightarrow \infty} |x_{k+1} - x_k| = 0$$

Finally:

$$\lim_{k \rightarrow \infty} x_{k+1} = \lim_{k \rightarrow \infty} x_k = \lim_{k \rightarrow \infty} \varphi(x_k) = \alpha$$

- Proved: If $\frac{\sqrt{2}-1}{2} < c < 1$, the algorithm converges!

If $c = 0.5$: Newton

$$x_{k+1} = \frac{1}{2} \left(x_k + \frac{2}{x_k} \right)$$

c can be different

Advanced way

Acceleration: $\varphi(\alpha) = \alpha$

$$x_{k+1} - \alpha = \varphi(x_k) - \varphi(\alpha) = \varphi'(\xi_k)(x_k - \alpha)$$

$$x_{k+2} - \alpha = \varphi(x_{k+1}) - \varphi(\alpha) = \varphi'(\xi_{k+1})(x_{k+1} - \alpha)$$

For the iteration converges:

$$\lim_{k \rightarrow \infty} \xi_{k+1} = \lim_{k \rightarrow \infty} \xi_k = \alpha$$

Thus:

$$\frac{x_{k+2} - \alpha}{x_{k+1} - \alpha} = \frac{x_{k+1} - \alpha}{x_k - \alpha}$$

Multiply:

$$(x_{k+2} - \alpha)(x_k - \alpha) = (x_{k+1} - \alpha)^2$$

Expand it:

$$x_{k+2}x_k - \alpha(x_{k+2} + x_k) = x_{k+1}^2 - 2\alpha x_{k+1}$$

Combine:

$$\alpha = \frac{x_{k+2}x_k - x_{k+1}^2}{x_{k+2} - 2x_{k+1} + x_k} = x_k - \frac{(x_{k+1} - x_k)^2}{x_{k+2} - 2x_{k+1} + x_k}$$

At last:

$$\alpha = x_k - \frac{(\varphi(x_k) - x_k)^2}{\varphi(\varphi(x_k)) - 2\varphi(x_k) + x_k}$$

New Iteration:

$$x_{k+1}^* = \psi(x_k) = x_k - \frac{(\varphi(x_k) - x_k)^2}{\varphi(\varphi(x_k)) - 2\varphi(x_k) + x_k}$$

Convergent order: ≥ 2 (Not going to prove it)

Algorithm setup

- Normal way:

$$x_{k+1} = cx_k + (1 - c)\frac{2}{x_k}$$

- Advanced way:

$$x_{k+1}^* = \psi(x_k) = x_k - \frac{(\varphi(x_k) - x_k)^2}{\varphi(\varphi(x_k)) - 2\varphi(x_k) + x_k}$$

where

